An Overview of Cloud Computing

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Abstract

As technology gets more and more sophisticated cloud computing has emerged as a possible solution for some companies as a way to out source their IT systems so that it doesn't need to be help on premise. In the beginning this paper covers a basic over view of how the cloud works and how it might be beneficial for some companies to adopt. Within the paper the history of the cloud is also covered to provide an in-depth idea of where the cloud came from. The paper then explains the structure and function of the cloud, debunking common misconceptions about its nature and highlighting the various services offered by providers such as AWS, Google, and Microsoft. The paper also discusses the different services that cloud providers give such as Software as a Service, Infrastructure as a service and platform as a service. After that the paper goes into detail explaining what it takes to protect cloud systems from attacks as well as goes over how cyber security works in the cloud. Then the paper also goes into the pros and cons that need to be looked at when deciding whether or not adopting the cloud might be good for a company. Finally, the paper goes over some various examples explaining why some companies might not want to go to the cloud. One of the examples being why nuclear powerplants decided against going to the cloud for their internal system.

Introduction

With the world of technology ever evolving it is important to make sure that you are aware of what new technologies have begun to come into play as the years go by. The cloud is something that has been gaining traction as it is a great way to outsource IT infrastructure as a business to save on costs.

The reason why this paper is being written is during the process of making a cloud system on AWS in the spring semester of 2024, I learned how to use a great deal of tools to create a system that worked together to create a well-rounded platform that monitored statistics from a website that I created and also from a Minecraft server. During the creation process, I learned how to use the tools and make them work together. Unfortunately, during that process, I did not get a chance to really take a deep dive into how the systems worked and why they worked, as I was up against a hard deadline to finish the project by the end of the semester so that it could be presented. That's why the intention of this paper focuses on how cloud technology works on a basic level. I will also provide a general understanding of how the cloud works to possibly help companies decide whether or not going to the cloud would be beneficial for them. The paper will begin by going over the history of the cloud as it is important to understand where it originated from. Next, I will be defining what the cloud even is. A common misconception is that the cloud is simply a cloud network in the sky where information magically gets stored, but in all reality, this is not the case at all. Next, I will go into how the cloud works and the different services that most cloud providers offer, along with an overview of how security is upheld when creating the cloud system.

Next, I will just give a basic pro and con list to define why a company may or may not choose to use the cloud in their systems and what the possible benefits of using it are. Finally, I will go over some examples of companies that don't use the cloud and the reasons why they don't, and also give an overview of why the cloud might not be the best option for some companies.

History

Over time many things can be traced back to the creation of the cloud but of those these are the ones that seemed like they made the biggest impact to me over the years of the development of the cloud. Everything in this section was pulled from the article written by Foote relating to the history of the cloud.

Timeline

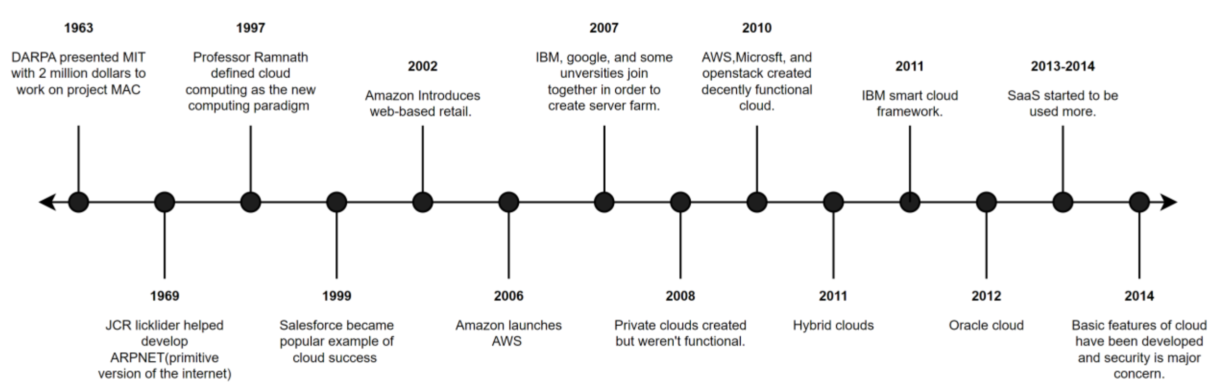

Figure 1 Cloud History Timeline

1963-DARPA

The idea behind Project Mac is pretty simple as the main goal of it was to allow two people to be working on a computer at the same time. While this doesn't sound that impressive for this date in time it was a big step as the computers that we are talking about were the very old mainframes. These were computers that had tapes running on them to provide them with the computational power that they needed to function. Additionally, at this point, a partition would have needed to be set up to make this work.

1969-ARPNET

The idea behind ARPNET was basically supposed to be what the internet is today. The main developer who worked on this was JCR Licklider, he was a psychologist and computer scientist who said that his vision was to make something called the "Intergalactic Computer Network" which was designed to allow people across the world to be connected to each other via computers. Which sounds pretty much exactly like what the internet is today.

2002- Amazon

At this point in the timeline, Amazon had become an online retailer that allowed third-party companies to sell their products to consumers to turn a profit. But it's also worth mentioning that in 1995 Amazon was originally an online book seller. As traction was gained by Amazon other large retail businesses started to join in to the new world that is online shopping.

2006-Amazon AWS

At this point, Amazon launched AWS to provide businesses that couldn't afford to uphold their own IT architecture, so Amazon said they would do it for them by introducing a couple of services, Amazon Mechanical Turk which provided cloud-based services and storage that clients could use. Then there was also EC2, which stands for, elastic compute cloud, which was designed to be a computer system hosted in the cloud and is still used in today's world but it is much more sophisticated.

2007- Server farm

At this point, Google, IBM, and some universities joined together to create a server farm that could be used for research and projects that required fast processing and huge amounts of data. There were many universities that joined this concept as they realized that it would be much cheaper to outsource their computing, and it also allowed them to get projects done much faster.

2011- Hybrid cloud

The idea behind a Hybrid cloud was that companies could use their existing servers that they already had on-premises along with the cloud servers that were available to them to enhance their systems. For example, if there were days when you knew that you would need more resources because you were going to have a higher flow of traffic on your network, then you could notify the cloud company of that information and then they would give you additional resources to work with. While this does sound like a pretty good deal during this time there were very few companies that could support this process as their systems weren’t compatible with the ones that the cloud providers were running.

2013-2014-SAAS started to gain traction

Usually, companies like to have full control over their systems which is why things like IAAS (infrastructure as a service) and PAAS (platform as a service) are usually the best option.

This allows the companies to control everything. But in some cases that much work isn't needed which is where SAAS (software as a service) comes into play. This option allowed companies to start up simple services quickly that were already provided by the cloud providers.

2014- Basics of the cloud are set up

At this point, the basic functionality of the cloud was set up, and the concern moved to security as all of an organization's data was now being outsourced to another company that wasn't internally within the organization. Since then cloud companies have been focusing a lot more on assuring that their systems are secure so that their customers can rest easy knowing that their data is protected.

Containers

While this is not included in the timeline that can be seen above, I feel like it is important to go over a little of the history for containers as they are how most systems where multiple things are running at the same time work. In 2004 there was of course Solaris Containers which was very primitive, it wasn't until 2013 that things like Docker came into play and since then there have been hundreds of tools that have been developed in order to start containers on a system. One of the most notable being Kubernetes, which works using nodes in order to make a complete system.

An Overview of the cloud

When you say cloud to most people in your everyday life the first thing that they will often think of is the thing that rain comes out of. However, for some weird reason if you add the word, the, in front of the word, cloud, then they know that you are talking about the thing that helps a lot of companies' function. Most people however don't even really know what the cloud, is at the end of the day and tend to believe that it is some magical device that somehow keeps the internet up and running. What people don't realize is that the cloud is made up of physical devices that are designed to provide users with services in order to help them build up their business infrastructure. Take AWS for example who has massive data centers all around the world in order to make sure that their clients get the best possible network services. Overall, the cloud can be defined as "it's a term used to describe a global network of servers, each with a unique function. The cloud is not a physical entity, but instead is a vast network of remote servers around the globe which are hooked together and meant to operate as a single ecosystem" (*What is the cloud - definition: Microsoft Azure*). Having all of these servers make up the internet allows users from all over the world to connect together in order to connect for various reasons whether that be business, gaming, politics, friendships, and many other reasons.

How the cloud works

On a basic level, it can be understood that the cloud is a group of servers that are all pooled together to allow people to use their resources to perform business operations. However, another important thing to note is that these aren't just simply a few servers that are doing this but are instead huge data centers all over the country and world that allow for the best possible outcome and the minimal amount of downtime. For example, if an entire region of data centers were to go down then all of the businesses relying on those servers would be transferred over to a different region and while the system would be behaving much slower it would not fully go down due to having data centers all over. Other functions of the cloud also should be mentioned. In an article titled *Cyberattacks and Security of Cloud Computing: A Complete Guideline* the vital features of the cloud were defined as “The five vital features of cloud computing include on-demand self-service, extensive access to the network, resource pooling, high-speed resiliency, and measurable services [3]. These benefits encourage large companies to dump their IT infrastructure into the cloud environment” (Dawood et al., 2023). All five of these different features are important to make sure that businesses can stay up and running no matter the circumstances. Take for example the resource pooling feature which is meant to help with the idea of automatic scaling some companies have a very unstable client base and because of that, they might need more or fewer resources. There is also the point of measurable service so that the cloud providers can understand how many resources their users are taking up to determine how to best charge them for the services. It is also worth mentioning though that there are different types of cloud services that are offered and depending on what an organization chooses this can change what the end user is responsible for. Those types of cloud services are Infrastructure as a Service, Platform as a Service, and Software as a Service. Depending on what kind of service users pick they have to be concerned about different things. See the chart below for an explanation. (The sections Infrastructure as a service, Platform as a service and Software as a service were pull from the article The shared responsibility model and SAAS, explained and the cyber security section relates to *What is cyber resilience?*)

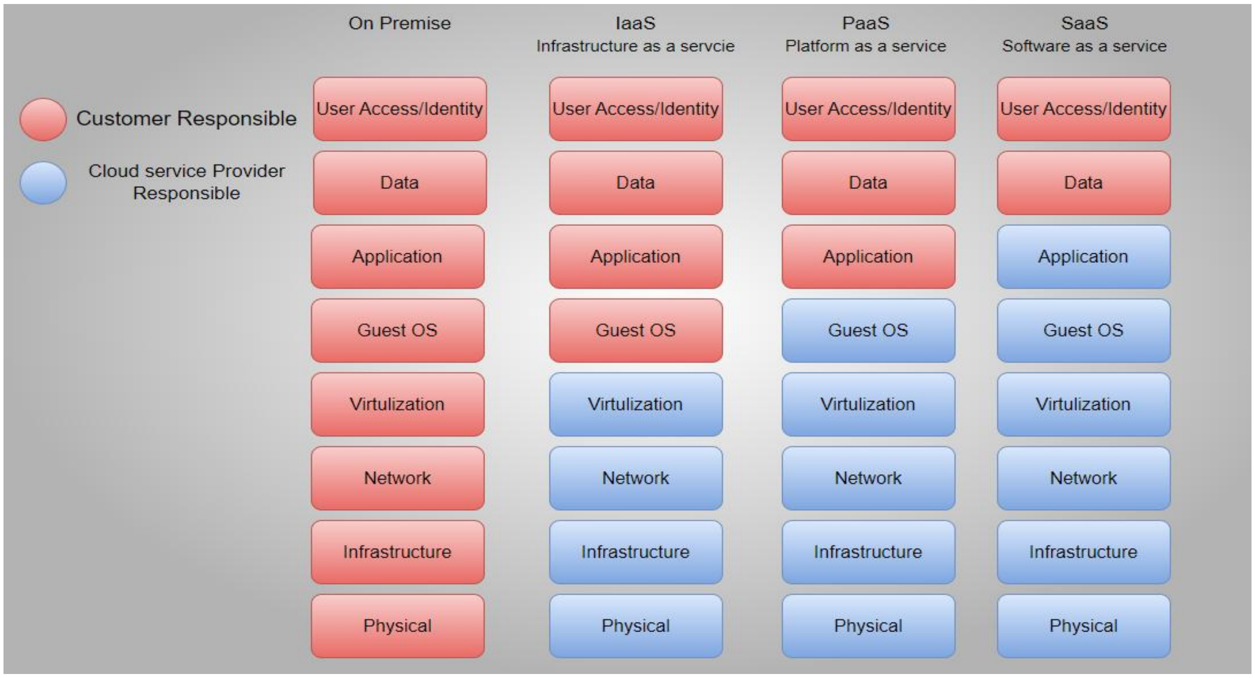

Figure 2 Cloud service tiers

Infrastructure as a Service

As can be seen by the figure above, IaaS is the most intensive on the user end when it comes to setting up a cloud system. But it Is much better when compared to On-Premises. While the users don't need to set up the physical aspect of their cloud system, they do need to set up everything else that is running. This means that they will be in charge of setting up the operating system that their servers are running on as well as creating the application that will be running on those servers. Additionally, the user will be in charge of setting up how their data is stored and how it is also protected. Finally, users will also be in charge of controlling who can access what in the system by setting up things like access controls.

Platform as a Service

PaaS is about the same as IaaS but in this case, the user is not in charge of setting up the OS that the server is going to run on, and they will instead be on the shoulders of the cloud service provider. However, users will be responsible for setting up the application, data, and access controls. Using this service allows clients to have more control over their system without having to worry about setting up all of the infrastructure as that labor is offloaded to the service providers.

Software as a Service

Finally, there is SaaS which is the simplest and easiest to manage out of all of the different service types that you can do with the cloud. In this type of service, you will likely be using one of the cloud-prebuilt applications to perform whatever function you need it to accomplish. You will however still be in charge of managing how data is stored as well as access controls for your system.

Cyber Security

Another thing to mention when it comes to cloud technology is the idea of cyber security. The reason why is that people start to get wary when the idea of something being hosted online gets mentioned because online is where most hacks happen. That being said there are many checks that are put into place within a cloud service to determine what and who can access the system. For example, access control lists can be put into place to manage who is and who isn't allowed to see certain things within a system. If say you don't want a specific user or group of users to see an AWS S3 bucket, then you can add an access control list that prohibits said parties from doing so.

Another thing that can be put into place is firewall rules on an AWS EC2 instance(virtual computer) or something similar in other cloud services which will allow you to control what kind of network traffic is allowed in and out of the instance depending on what that instance is trying to accomplish. Another thing to mention about cloud services is the idea of Cyber Resilience. In an article that I found this was the definition given to this concept "Cyber resilience is a concept that brings business continuity, information systems security, and organizational resilience together. The concept describes the ability to continue delivering intended outcomes despite experiencing challenging cyber events, such as cyberattacks, natural disasters, or economic slumps. A measured level of information security proficiency and resilience affects how well an organization can continue business operations with little to no downtime" (*What is cyber resilience?*,2024). This concept is something that cloud providers built their platforms around because of the idea that time is of course money and the longer that a company is not in operation the more money they are losing if they cannot function. So, if a cyber-attack were to happen on a cloud service provider there are procedures that are put into place to mitigate the attack and keep as many businesses up and running as possible.

Additionally, it is a good idea to understand the different attacks that a cloud system might undergo, and to be honest they are pretty similar to if the equipment was on-premises. In an article this quote was stated “Cloud computing faces several typical cloud risks, such as data abuse, malicious insider, insecure interfaces, access points, shared technological problems, loss of data, and hijacking.” (Dawood et al., 2023). While these attacks are very serious if they are to occur cloud services understand that they can happen and because of that have put in checks to make sure that the businesses that they service are as protected as they can be to mitigate the impact of an attack.

Pros and Cons

To best describe and give a general overview of what the cloud is, it is important to also go over the pros and cons. This next section talks about why some companies might want to or not want to use the cloud.

Pros	Cons
Cost Effective One of the major benefits when it comes to using the cloud is the fact that it can be very cost-effective if you don’t want to pay the overhead of having all the equipment needed on-premises. The reason for this is that you one won’t have to buy all of the equipment and two you won't have to maintain all of it and pay the labor overhead to do so.	**Understanding cost** One of the downfalls of using a cloud provider is the fact that the cost of everything can oftentimes be confusing and takes a legal team to understand. To make sure you are getting the best bang for your buck you will need to research to understand what the costs all mean.
Defend against disaster Since there are so many data centers across the world it is less likely that your	**Limited Control** As can be seen in the figure above you have no control over what equipment your

data is going to be lost in the event of something like a fire because said data is redundant across data centers which leads to fewer incidents of data loss.	system is running on. Meaning that if your system needs a certain type of hardware in order to run properly you might run into some issues.
Maximize uptime If your server were to go down and it is in your location, then you are going to have downtime. However, using a cloud service minimizes downtime. If a server goes down, then it just simply will move all of your stuff to a new one and be back up within minutes.	**Vendor Lock-In** When switching to the cloud there might be complications with the vendor which can lead to unneeded data vulnerabilities that could hurt your company in the long run if an attack were to happen.
Scalability Usually in order to scale a server a company will sometimes have to wait to get the part to do so because of a procurement process. But when using the cloud that isn't an issue because if you need more resources then all you have to do is ask and they will give you more.	**Internet Reliance** While this is a minute issue it is obvious that to use your cloud service you need to have an internet connection so if you lose that then you will no longer have access to any of the data that is hosted by the cloud provider.
Space In order to have a business you need physical property and property is expensive, especially as it gets bigger. So, when you add a server rack the need for space does increases. Going to the cloud can allow you to either go with a smaller physical property or use the space where	

the server would have went for something else.	

Why or why not use cloud?

Cloud is a great thing for most companies as it provides an easy way to get your business up and running without having to do any of the labor-intensive setup with all of the physical hardware. But in some cases, going to the cloud might not be the most suitable option for a business for various reasons. When a company is small it can be assumed that they most likely won't have the most money to work with meaning that they can't pay for the overhead of buying the servers to run their business. In cases like these going to the cloud is a great option as it is set up as a pay-as-you-go situation. Additionally, their different service types allow everyone no matter what their technical skills are to use the cloud. Bigger corporations, however, might benefit from having their own on-premise servers if it better serves their business operations. But this would of course require that they have a team of IT specialists that can monitor the server and do any maintenance work that might be required during the life span of the company. However, in some cases going to the cloud might not be a viable option for some companies because they might have higher security in place.

Take nuclear power for example, which can be considered a governmental entity. Nuclear power is a special case as there are two different sides to how business gets run at the nuclear plant. The first side that I want to talk about is the business side of nuclear as it doesn't deal with anything specifically connected to the reactor at the plant. When I say business I mean things like human resources, marketing, payroll, accounting, and of course other normal business functions. All of these functions can be done by using the cloud as there are no special procedures that need to be followed to perform them. But another concern comes with the idea of security as doing things like payroll often includes very sensitive information about a person. But by using cloud security features such as access controls, admins can control who sees what on the website/system to enforce security to the best of their ability. Using the cloud for nuclear power is also a way to maximize the output of energy that is going to the grid as they don't have to provide power for overhead to run the servers.

The nuclear side of the business is a whole different beast as there is much more security that needs to be implemented to protect the plant from any possible attacks. For this reason, going to the cloud has not been a viable option. The reason why is that employees have to be extremely specific with how they do their work. For example, in order to do anything on any piece of equipment at least two employees need to be present and they need to confirm with each other after everything that they do to make sure that they don't accidentally do something that might break the plant in any way. Additionally, the servers that the plant is running on need to be of a specific specification meaning that the components within the servers need to be a specific brand. To even get the hardware for the servers if they need to be replaced employees need to go through a procurement process that is rigorous and can sometimes take months to deliver a single part. Finally, the last reason that I want to go over is what I believe is the main reason why the cloud is not used in nuclear power. One of the cons that I had talked about earlier was the idea that to use cloud services you need to have a stable internet connection. If you were to lose that connection, then you are going to lose your connection to the data that might be vital to perform main business functions. In nuclear the operators that are up in the control room need to be monitoring the plant at all times and look at hundreds upon hundreds of data points that all tell them something different about what's going on in the reactor, or what's going on with the turbines that create the power that the plant sends out to the grid. These points also all put out real-time data that the operators look at and for safety reasons, they need to be able to see the real-time data at all times. So, if all of the points are hosted on a cloud server and the plant happens to lose internet then they will no longer be able to see the real-time data from the plant which could be detrimental depending on the state that the reactor is in. For this reason, the servers that handle all of the data points that the plant spits out need to be on-premise to make sure that there is the least amount of downtime possible when it comes to monitoring the plant. Another small thing to note is that all of the sensors that are sending out the data in the first place are connected to the reactor directly on premises meaning that it doesn't make a whole lot of sense to have your data sent out to the cloud and then be brought back down from the cloud when you can skip the middle man and just keep all of the data on-premise.

Conclusion

While it is true that going to the cloud is a great option for most companies there are some instances where it may not be the best option. So that being said making sure to analyze your company to determine if the cloud is right for you is important. Some things that might need to be considered when determining whether or not to switch to the cloud are if they can provide the right amount of security for you if they can provide you with the type of devices that you need to perform business functions, and if it is cost-effective for your company to switch to the cloud. All of these different questions need to be answered to decide whether or not the cloud will provide you with the best possible service for your systems when compared to on-premises systems.

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